

Power BI Report Project: Skills Checklist

Use this checklist to track your progress through every required skill for building, publishing, and presenting a professional Power BI report. Check each item off as you complete it.

Section 1 — Connecting & Cleaning Data

☐ Import multiple data sources into Power Query

Connect to all provided data files — typically one or more Excel workbooks and at least one text file. In Power Query, load each sheet or file as its own named query. Rename queries clearly so they are easy to identify throughout the project.

Tip: Use Get Data → Excel Workbook for spreadsheets and Get Data → Text/CSV for flat files. Load each sheet as a separate query rather than combining them at import time.

☐ Fix data types and date formatting

Inspect every column's data type after import. Dates stored as Excel serial numbers must be changed to the Date type in Power Query. Dates split across Year, Month, and Day columns must be merged into a single Date column using a custom column formula.

Tip: To merge split date columns, use Add Column → Custom Column and enter: Date.From(#date([Year],[Month],[Day])). Then remove the original three columns.

☐ Handle errors, nulls, and invalid values

Scan each column for error values (such as #N/A), blank cells, and clearly invalid entries (such as negative prices or non-numeric IDs). Use Replace Values, Replace Errors, Remove Rows, or conditional columns in Power Query to address each issue. Document what you changed and why.

Tip: Before fixing, note how many rows are affected. After fixing, use Column Quality in the View tab to confirm that error and empty percentages have dropped to zero.

☐ Filter data to your assigned scope

The dataset may cover multiple regions, stores, or time periods. Use your project brief to identify which records belong to your scope, then apply filters in Power Query so that only relevant rows are loaded into your model. This keeps visuals accurate and model performance efficient.

Tip: Join your store or location dimension table to the management or hierarchy table first — this tells you which store IDs belong to your region before you filter the main fact table.

☐ Normalize and combine data from two or more fact tables

Sales data is often split across multiple sheets with different column names or date formats — for example, in-store transactions in one sheet and online orders in another. Normalize column names and date formats in Power Query so both tables align, then either append them into a unified fact table or manage them as two related tables in the model.

Tip: Before appending, set both tables' date columns to the same Date type and confirm that numeric amount columns use the same name and format. Appending tables with mismatched column names creates null-filled columns.

☐ Enrich sales data by merging with dimension tables

Use Merge Queries (equivalent to a JOIN) to add descriptive fields from dimension tables into your sales rows. Common merges: joining a Products table on a product ID to add category and name fields; joining a reference text file on a product ID to add attributes such as author, brand, or description.

Tip: Set the join kind to Left Outer so all sales rows are kept, even if the dimension table has no matching entry. Expand only the columns you need to avoid inflating the query with unnecessary fields.

Section 2 — Building the Data Model

☐ Create a Date / Calendar table

Build a continuous date table that spans the full date range in your data. In DAX, use `CALENDAR(DATE(start year,1,1), DATE(end year,12,31))`. Add calculated columns for Year, Month Number, Month Name, Quarter, and Weekday. Mark the table as a date table in Model View so Power BI enables time intelligence functions.

Tip: Month names will sort alphabetically by default. Fix this by setting the Sort by Column for Month Name to Month Number in the Column Tools ribbon.

☐ Define table relationships in Model View

In Model View, draw relationships between your dimension tables and fact tables. Most relationships should be one-to-many with a single filter direction: the dimension (one side) filters the fact table (many side). Connect your Date table to the date column in each fact table. Connect Products to sales using the product ID. Connect Store Locations to in-store sales using the store ID.

Tip: After creating relationships, test them by placing a dimension column (such as Category) on a visual — if it does not filter the sales data, the relationship is missing, set in the wrong direction, or uses mismatched column types.

☐ Write explicit DAX measures

Create named DAX measures rather than relying on Power BI's automatic implicit aggregations. At minimum write a Total Sales measure using `SUM()` that references your fact table's revenue column. Use this measure in every visual. Additional useful measures include year-to-date totals, prior-year comparisons, and ranked counts.

Tip: Explicit measures appear in the Fields pane with a calculator icon and can be reused across every visual and page. Implicit aggregations (drag-and-drop raw columns) cannot be reused and are harder to audit. Replace all implicit aggregations with your explicit measure once it is written.

Section 3 — Building Report Visuals (4 required)

☐ Line chart — sales trend over time

Add a Line Chart visual. Set the X-axis to the Date hierarchy from your Calendar table (Year → Quarter → Month) and the Y-axis to your Total Sales measure. The chart must cover the full date range in the data and allow users to drill down to the monthly level by clicking the drill-down arrows.

Tip: If the X-axis groups incorrectly or shows gaps, confirm that your Calendar table is marked as a date table and that its relationship to the fact table is active and correctly typed.

☐ Column or bar chart — sales performance by category

Add a Clustered Column Chart or Bar Chart. Set the axis to the Category field from your Products or dimension table and the value to your Total Sales measure. Sort the chart in descending order by sales so the highest-revenue category appears first. Confirm that cleaned category values appear — no error or null categories should be visible.

Tip: To sort descending: click the ellipsis (...) on the visual → Sort by → Total Sales → Sort descending. If the sort option is grayed out, the visual may still be using an implicit aggregation rather than your measure.

☐ Pie or donut chart — relative sales by state

Add a Pie Chart or Donut Chart. Set the Legend to your State dimension — this should include both in-store states (pulled from the Store Locations table) and online ship-to states (pulled from the online orders table). Set the Values field to your Total Sales measure. Confirm that all states in your scope appear as slices.

Tip: If in-store and online states come from two different columns in two different tables, you may need to create a unified State dimension or use a bridge approach so the same slicer and visual can cover both sources.

☐	Table or matrix visual — top-selling items with detail Add a Table or Matrix visual showing your top-selling products. Include the item name, a descriptive attribute joined from your reference file (such as author, brand, or endorser), and the Total Sales measure. Apply a visual-level filter to exclude any category that should not appear (for example, textbooks or media). Sort by Total Sales descending. <i>Tip: To show only the top N items, add a visual-level filter: drag Total Sales into the Filters on this visual pane, set filter type to Top N, enter the number of items, and drag Total Sales into the By value field as well.</i>
☐	Add descriptive page headers to every report page Every page in your report (minimum two pages are required) must have a clearly readable text header that describes the content of that page. Insert a Text Box from the Insert tab, or use a formatted Shape with text. The header should be visible at 100% zoom without scrolling and must be written in plain language — not a default label. <i>Tip: Ask yourself: if a store manager opened this report without any prior briefing, would they immediately understand what each page shows? If not, the header needs to be clearer or more descriptive.</i>
☐	Add at least one interactive slicer Add a Slicer visual to enable viewers to filter the report interactively. Good options for most sales reports are a Year slicer, a State or Region slicer, or a Category slicer. Test that selecting a value in the slicer correctly updates all other visuals on the same page. You must demonstrate a live filter interaction during your video recording. <i>Tip: If a visual is not responding to the slicer, open the Format tab → Edit Interactions and check that the slicer is set to filter (not highlight or none) for that visual.</i>
☐	Format and label all visuals clearly Every chart must have a meaningful title, clearly labeled axes, and clean legends. Replace all default labels such as 'Sum of Sale Amount' with plain English equivalents such as 'Total Sales (\$)'. Remove unnecessary gridlines or chart elements that clutter the view. Use consistent colors and fonts across all pages. <i>Tip: Work through every visual using the Format pane: update the Visual title, rename X-axis and Y-axis titles, and check data labels if enabled. A common mistake is leaving the default title turned off entirely — every visual should have one.</i>

Section 4 — Publishing the Report

☐	Save the .pbix file with the correct filename Save your Power BI report as a .pbix file using the naming convention specified in your project brief (typically YourLastName_projectname.pbix). Save the file to your designated project folder. Power BI Desktop does not auto-save — get into the habit of saving with Ctrl+S frequently throughout the week. <i>Tip: Keep your .pbix file inside your project repository folder so that saving and committing to GitHub happen in the same location.</i>
☐	Publish the final report to Power BI Service When your report is complete, use the Publish button in the Home tab of Power BI Desktop to upload it to the shared class workspace in Power BI Service. Confirm the workspace name before publishing. Check that the report opens correctly in the browser after publishing — visuals should render and slicers should function. <i>Tip: If publishing fails with a credential error, go to File → Options and Settings → Data Source Settings and update the credentials for your data sources before re-publishing.</i>

Section 5 — GitHub Repository

☐ Create and clone the project repository

Create a new GitHub repository using the name specified in your project brief. Clone it to your local project folder before beginning any work so that all files are tracked from the start. Confirm that the local folder and the remote repository are in sync before you save any project files.

Tip: Run `git status` in your terminal after cloning to confirm the repository is connected. Then save your .pbix file into that folder so your very first save is already version-controlled.

☐ Commit daily with meaningful commit messages

Make at least one commit every working day during the project week. Each commit message should clearly describe what changed — not just 'update' or 'work in progress'. Your commit history is evidence of steady, organized progress and is part of your project evaluation.

Tip: Good commit messages are specific and past-tense: 'Added line chart for monthly sales trend', 'Fixed date serial number format in Power Query', 'Published final report to class workspace'. Avoid vague messages like 'changes' or 'stuff'.

☐ Write a README.md for the repository

Add a README.md file to your repository that briefly describes the project: what it analyzes, what data sources were used, what the report contains, and a link to your video recording. The README is the first thing a visitor sees when they open your repository and should make the project's purpose immediately clear.

Tip: Structure your README with at least three short sections: Project Overview (1–2 sentences), Data Sources (list the files used), and Video Walkthrough (paste the link). You do not need to write paragraphs — clear, concise bullet points work well.

Section 6 — Video Recording & In-Class Presentation

☐ Record a 10-minute walkthrough video

Record a screen-share video walking through your completed Power BI report. Cover each of the four required visuals one at a time, explaining what the chart shows and what insight it provides. Demonstrate at least one live filter interaction using a slicer. Include a brief discussion of how you approached data cleaning and modeling — what issues you found, what decisions you made, and what you learned.

Tip: Structure your 10 minutes: ~5 min on the four visuals and insights, ~3 min on data cleaning and modeling decisions, ~2 min on challenges or things you would improve. Record a rough draft first to check your timing before doing a final take.

☐ Deliver a 5-minute in-class presentation

Present your report live to your classmates as if you are addressing the organization's regional management team. Choose one or two visuals to focus on and lead with the insight that matters most to the business audience — not the technical steps you took. Practice your timing in advance: five minutes passes faster than you expect. Have your report open and your screen share ready before your turn.

Tip: Avoid walking through every visual equally. Pick your strongest insight and build your presentation around it. End with one clear recommendation: what should the team do differently based on what the data shows?

Common Data Quirks to Watch For

Sales datasets frequently contain the issues listed below. These are not mistakes in the data files — they are realistic data quality challenges you are expected to find and resolve in Power Query before building any visuals.

Issue to find	How to resolve it in Power Query
Dates stored as Excel serial numbers	The date column contains integers like 44562 instead of readable dates. In Power Query, click the column header → Change Type → Date. Power BI will convert the serial to the correct calendar date automatically.
Date split into Year, Month, Day columns	Three separate numeric columns instead of one date. Add a Custom Column using: <code>Date.From(#date([Year],[Month],[Day]))</code> . Then delete the original three columns and set the new column type to Date.
Error values in a category column (#N/A, #REF!, etc.)	Error values in a category or dimension column will appear as a separate unlabeled group in every visual. Use Transform → Replace Errors to substitute a label like 'Uncategorized', or use Remove Errors to drop those rows — whichever is appropriate for your analysis.
Floating-point precision in numeric amounts	Sale amounts may contain values like 205.80000000000001 due to floating-point arithmetic in Excel. In Power Query, add a custom column using <code>Number.Round([Sale Amount], 2)</code> to round all values to two decimal places.
Dataset covers more than your assigned scope	The provided data may include multiple regions, states, or store groups beyond the ones you are assigned to analyze. Join your dimension tables to identify which store IDs, states, or categories are in your scope, then filter the fact table rows before loading data into the model.
Two fact tables with incompatible formats	In-store and online sales are often provided as separate sheets with different column names and date structures. Rename columns and normalize date formats in Power Query before appending or relating the two tables so that one measure covers both sources consistently.
Text reference file with a non-standard delimiter	Reference files (such as author or product lists) may use a pipe or tab as a delimiter instead of a comma. When importing with Get Data → Text/CSV, change the Delimiter dropdown from Comma to Custom and type the correct character. Check for header rows at the top of the file that need to be skipped.